

Characterizing the Security of Github CI Workflows

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Characterizing the Security of Github CI Workflows

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Continuous integration and deployment (CI/CD) has revolutionized software development and maintenance. Commercial CI/CD platforms provide services for specifying and running CI/CD actions. However, they present a security risk in their own right, given their privileged access to secrets, infrastructure, and ability to fetch and execute arbitrary code.

In this paper, we study the security of the newly popular GitHub CI platform. We first identify four fundamental security properties that must hold for any CI/CD system: Admittance Control, Execution Control, Code Control, and Access to Secrets. We then examine if GitHub CI enforces these properties in comparison with the other five popular CI/CD platforms. We perform a comprehensive analysis of 447,238 workflows spanning 213,854 GitHub repositories. We made several disturbing observations. Our analysis shows that 99.8% of workflows are overprivileged and have read-write access (instead of read-only) to the repository. In addition, 23.7% of workflows are triggerable by a pull_request and use code from the underlying repository. An attacker can exploit these workflows and execute arbitrary code as part of the workflow. Due to the modular nature of workflows, we find that 99.7% of repositories in our dataset execute some externally developed plugin, called "Actions" , for various purposes. We found that 97% of repositories execute at least one Action that does not originate with a verified creator, and 18% of repositories

in our dataset execute at least one Action with missing security updates. These represent potential attack vectors that can be used to compromise the execution of workflows, consequently leading to supply chain attacks. This work highlights the systemic risks inherent in CI/CD platforms like GitHub CI; we also present our own Github action, GWChecker, which functions as an early warning system for bad practices that violate the identified security properties.

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